Reg. No : | 2 | 1 | 1 | 6 | 2 | 4 | 1 | 7 | 0 | 1 | 0 | 0 | 6 |

Question Paper Code: 051171

B.E./B.Tech DEGREE EXAMINATION APRIL/MAY - 2026

Fourth Semester

Computer Science and Design

AI23231 PRINCIPLES OF ARTIFICIAL INTELLIGENCE

[Common to: IT, AIML and AIDS]

(Regulations 2023)

Time : Three Hours | Maximum : 100 Marks

Answer all Questions

PART A (10x2=20 Marks)

1.	Define Artificial Intelligence.	[CO1]	[K1]
2.	Differentiate between goal-based and utility-based agents.	[CO1]	[K 2]
3.	Define heuristic function.	[CO2]	[K 1]
4.	Compare BFS and DFS.	[CO2]	[K 2]
5.	Define minimax algorithm.	[CO3]	[K1]
6.	State adversarial search with an example.	[CO3]	[K2]
7.	Define logic programming.	[CO4]	[K1]
8.	Differentiate forward reasoning and backward reasoning.	[CO4]	[K 2]
9.	State Baye's theorem.	[CO5]	[K 2]
10.	Define fuzzy logic.	[CO5]	[K2]

PART B (5x13= 65 Marks)

11.	a.	Explain the concept of state space search to model the Tic-Tac-Toe problem and analyze the role of intelligent agents in solving it.	[CO1]	[K 2]
		[OR]		
	b.	Explain the structure of intelligent agents and describe goal-based, utility-based, and learning agents. Also, discuss how the nature of the environment influences their behavior.	[CO1]	[K2]

12.	a.	Apply Breadth First Search and A* search to a suitable problem and analyze their performance in terms of completeness, optimality, and time/space complexity. Compare the effectiveness of heuristic and uninformed search strategies.	[CO2]	[K3]
		[OR]		
	b.	Apply heuristic search techniques to solve an optimization problem and analyze the performance of Greedy Best-First Search and Hill Climbing. Discuss their limitations and possible improvements.	[CO2]	[K3]
13.	a.	Apply the minimax search procedure to a two-player game and analyze how optimal decisions are made. Illustrate with a suitable game tree and explain the step-by-step evaluation.	[CO3]	[K4]
		[OR]		
	b.	Evaluate the effectiveness of iterative deepening search in game playing. Apply it to a suitable example and analyze.	[CO3]	[K5]
14.	a.	Explain forward reasoning and backward reasoning with example.	[CO4]	[K4]
		[OR]		
	b.	Develop a logic programming model to solve a real-world problem and include rule-based representation for the model.	[CO4]	[K6]
15.	a.	Analyze the role of statistical reasoning in Artificial Intelligence by applying Bayesian Networks to a real-world problem.	[CO5]	[K5]
		[OR]		
	b.	Explain Dempster-Shafer Theory with example.	[CO5]	[K6]

PART C (1x15=15 Marks)

16.	a.	Assess the effectiveness of constraint satisfaction techniques in solving complex real-world problems. Evaluate different strategies and justify their limitations and strengths.		[K5]
		[OR]		
	b.	Design an intelligent agent for a real-world application by integrating search techniques and learning mechanisms. Develop its architecture and explain how it adapts to dynamic environments.		[K6]